

The HydraChain platform provides the following components in a blockchain.

- **Transaction cost:** This is the cost of executing computational steps in a transaction. In HydraChain, the cost is calculated on the basis of requirements.
- **Gas limits:** These are basic units for transaction executed in a blockchain. Every transaction is sub-divided into 'opcode' and every opcode has specific gas units based on the opcode type.
- **Genesis allocation:** This is related to the hashing and mining capacity of the blockchain.
- **Block time:** This is the time delay between mining or validation of two blocks.

Official website: <http://www.brainbot.com/projects.html>

Latest version: 0.3.0

MultiChain

MultiChain is an open source blockchain platform that enables user based networks to supposedly perform 1000 financial transactions per second. It enables enterprises to build and implement blockchain applications with speed. It is specially designed for customers with requirements similar to what Bitcoin offers but with special features for corporate entities. With MultiChain, banks can even build their own blockchains. It offers users 45 customisable parameters and changeable protocol-level restrictions.

The following are some of the unique features of MultiChain:

- 1000 transactions per second.
- Dynamic control over connections, sends and receives transactions, asset creation, streams and blocks.
- Cross-chain applications and users can issue millions of assets on a blockchain.
- Highly developer friendly and fully customisable.
- Extends the Bitcoin protocol, formats and APIs including full multi-signature support.

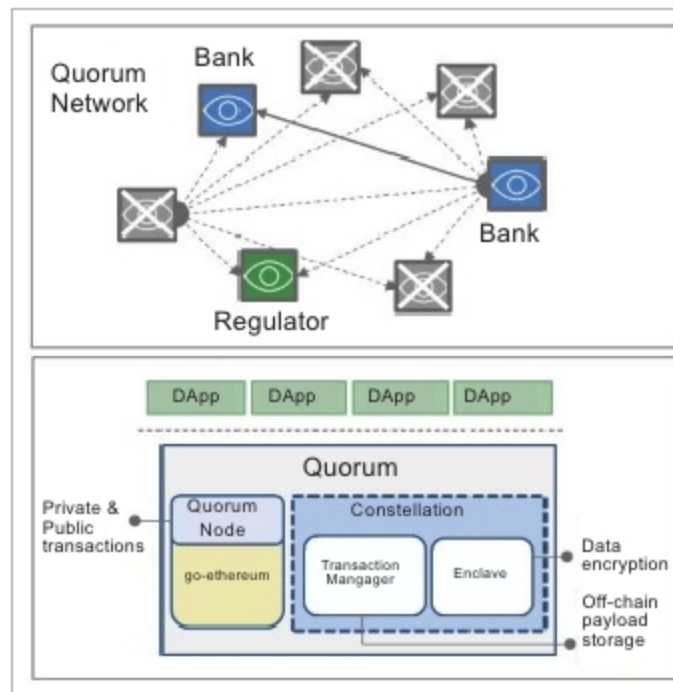


Figure 4: Quorum network architecture

- Ideal for data sharing, encryption archiving and time stamping.

Official website:

<https://www.multichain.com/>

Latest version: 1.0.4

Openchain

Openchain is a distributed ledger based on open source technology and is specially designed for organisations to issue and manage digital assets in a highly scalable, secured and robust manner. Openchain doesn't make use of the concept of blocks, and transactions are directly chained to one another. Transactions get linked to the chain at the same time that they are submitted to the network.

This is how it works:

- There is a single authority for validating transactions for every Openchain instance.
- Rather than have one single central ledger, every organisation has complete control over its respective instance and every instance connects with each other.
- Different transactions are validated by different authorities, depending on the exchange of assets.
- Every issuer of an asset has full control over the transactions. There are three modules of the Openchain blockchain.
 - **Storage engines:** These are the core components, and their primary task is to store

transaction chains and records. Modules include Openchain.Sqlite, Openchain.SqlServer, Openchain.MongoDB, etc

- Validation engines
- Anchoring media

Official website:

<https://www.openchain.org/>

Latest version: 0.7

Elements

Elements, being an open source collaborative project, was designed to bring advanced technological improvements to Bitcoin technology. It is a blockchain based digital currency that aims to revolutionise the loyalty rewards industry. It offers a unique platform to multiple merchants to leverage the underlying cryptocurrency technology to boost sales, visibility and customer satisfaction.

It operates on the Proof-of-Work (POW) mining algorithm with the X11-changed hashing algorithm as its POW. It provides a universal loyalty currency that can be adapted by various merchants and used by customers as per their requirements.

Components of the Elements blockchain are:

- Asset issuance
- Confidential transactions
- Segregated witnesses
- Relative time clock
- SCHNORR signatures
- New opcodes
- Signature covers value
- Deterministic pegs
- Signed blocks
- Bitmask Sighash, which is under development

Official website:

<https://elementsproject.org/>

Latest version: 0.14.1

Quorum

Quorum is an open source extension to Ethereum created by J.P. Morgan. It supports transaction and contract privacy. It was designed especially for applications requiring a high speed and throughput in terms of processing private transactions and a permissioned group of known participants.